

E01 Practicing with Stoichiometric Tables and Changes in Volumetric Flowrate

The gas phase reaction $A + 2B \rightarrow C$ is carried out isothermally and isobarically in a fluidized CSTR with no pressure drop. The feed is stoichiometric in A and B and no C is fed. The pressure, temperature, and volumetric flowrate of the feed gas are 6 atm, 170°C, and 50 L/min respectively. The rate constant $k_A = 53 \text{ mol/kg catalyst} \cdot \text{min} \cdot \text{atm}^3$ at 300 K, and the activation energy is 80 kJ/mol. Complete the stoichiometric table and write the rate as a function of conversion.

Decision Guide: Stoichiometric Tables and Changes in Volumetric Flowrate

Goal

Determine:

1. The stoichiometric table.
2. The concentration of each species as a function of conversion.
3. The reaction rate as a function of conversion.

Step 1: Identify the Reaction Conditions

Ask:

Gas phase or liquid phase?

→ Gas phase

Isothermal?

→ Yes

Isobaric?

→ Yes

Pressure drop?

→ No

Because the reaction occurs in the gas phase, volumetric flowrate will generally change as conversion changes.

Step 2: Choose a Basis Species and Interpret the Feed Composition

For stoichiometric-table problems, the next step is to determine which reactant will be used as the basis for the calculations. The basis species should be **the limiting reactant**. The problem states that the feed is **stoichiometric in A and B**. This means that A and B enter the reactor in exactly the proportions required by the balanced chemical equation.

For the reaction



two moles of B are required for every mole of A. Therefore:

For every 1 mole of A fed,

2 moles of B are fed.

No C is present in the feed.

This information allows us to determine the feed-composition parameters needed for the stoichiometric table:

$$\Theta_A = 1$$

$$\Theta_B = 2$$

$$\Theta_C = 0$$

The purpose of this step is to establish the starting composition of the reactor feed before calculating how conversion changes the amounts of each species.

Step 3: Calculate δ

Use:

$$\delta = \frac{d}{a} + \frac{c}{a} - \frac{b}{a} - 1$$



you would then substitute:

$$\delta = \frac{1}{1} + \frac{0}{1} - \frac{2}{1} - 1 = -2$$

which indicates that the total number of moles decreases as the reaction proceeds.

CBE 3610 Reactor Design Handbook Equation (50)

Step 4: Calculate ϵ

Use:

$$\epsilon = y_{A0}\delta$$

CBE 3610 Reactor Design Handbook Equation (53)

This parameter accounts for the effect of reaction on reactor expansion or contraction.

Step 5: Write the Volumetric Flowrate Expression

Because the problem is:

Gas phase

Isothermal

Isobaric

with

No pressure drop

use:

$$v = v_0(1 + \epsilon X)$$

which follows from the general expression.

CBE 3610 Reactor Design Handbook Equation (54)

This gives the volumetric flowrate as a function of conversion.

Step 6: Determine the Feed Concentration

The problem gives:

- Pressure
- Temperature
- Volumetric flowrate

Use the ideal-gas concentration relationship:

$$C_A = \frac{P_A}{RT} = \frac{y_A P_{tot}}{RT}$$

CBE 3610 Reactor Design Handbook Equation (1)

to determine:

$$C_{A0}$$

Step 7: Write Concentrations as Functions of Conversion

Use:

$$C_j = C_{A0} \frac{\theta_j + \nu_j X}{1 + \epsilon X}$$

because:

Gas phase

Isothermal

Isobaric

No pressure drop

allows the pressure and temperature ratios to simplify.

CBE 3610 Reactor Design Handbook Equation (55)

Determine:

$$\begin{aligned} &C_A(X) \\ &C_B(X) \\ &C_C(X) \end{aligned}$$

Step 8: Calculate the Rate Constant at Reaction Temperature

The problem provides:

- k at 300 K
- Activation energy
- Reactor temperature

Use:

$$k(T) = k(T_{ref}) \left[\frac{E}{R} \left(\frac{1}{T_{ref}} - \frac{1}{T} \right) \right]$$

CBE 3610 Reactor Design Handbook Equation (49)

to determine the rate constant at the reactor temperature.

Step 9: Write the Rate Law in Terms of Conversion

Begin with:

$$-r_A = [k(T)] \cdot [f(C_A, C_B)]$$

CBE 3610 Reactor Design Handbook Equation (44)

Substitute:

- $k(T)$
- $C_A(X)$
- $C_B(X)$

to obtain:

$$-r_A = f(X)$$

This is the final goal of the problem.

Final Answer Checklist

- ✓ Determined Θ_B and Θ_C
- ✓ Constructed the stoichiometric table
- ✓ Calculated δ using Eq. (50)
- ✓ Calculated ϵ using Eq. (53)
- ✓ Determined $v(X)$ using Eq. (54)
- ✓ Calculated concentrations using Eq. (55)
- ✓ Calculated $k(T)$ using Eq. (49)
- ✓ Written the rate as a function of conversion

CBE 3610 Reactor Design Handbook Equations Needed

Eq. (1) Ideal-Gas Concentration

Eq. (44) General Rate Law

Eq. (49) Temperature Dependence of k

Eq. (50) Change in Moles Parameter, δ

Eq. (53) Mole Change Parameter, ε

Eq. (54) Volumetric Flowrate as a Function of Conversion

Eq. (55) Concentration as a Function of Conversion

Key Insight

Reaction Changes Total Moles

↓

Total Moles Change v

↓

v Changes Concentration

↓

Concentration Changes Rate

For gas-phase reactions, writing the stoichiometric table is not just bookkeeping—it is the key step that allows you to determine how conversion influences concentration and reaction rate.